

Multiple Choice (10 marks)

1. Which of the following statements about lipids is correct?
 - (a) Lipids are the primary source of energy for cellular processes.
 - (b) Lipids are soluble in water but insoluble in organic solvents.
 - (c) Phospholipids are water soluble.
 - (d) Steroids are lipids that consist of glycerol and fatty acids.
 - (e) Lipids make up most cell surface receptors.
2. The theory of evolution is most accurately described as
 - (a) a universally accepted fact about the origin and development of all species.
 - (b) a hypothesis in the process of being tested and verified.
 - (c) a disproven theory about the development of species over time.
 - (d) a speculation about possible changes in populations over time.
 - (e) an opinion that some scientists hold about how living things change over time.
3. How can one demonstrate that amino acids are used to build proteins in cells, while the proteins already in the cell are used in catabolism?
 - (a) By observing the cell under a microscope
 - (b) Using fluorescent tagging of proteins
 - (c) Tracking changes in cell size and shape
 - (d) Autoradiography with labelled amino acids
 - (e) Through protein synthesis
4. Which of the following statements about mitochondria and chloroplasts is generally true?
 - (a) Plants, animals, and fungi all have both chloroplasts and mitochondria.
 - (b) Plants and fungi have chloroplasts but no mitochondria; animals have only mitochondria.
 - (c) Animals have both chloroplasts and mitochondria; plants and fungi have only chloroplasts.
 - (d) Plants and animals have chloroplasts but no mitochondria; fungi have only mitochondria.
 - (e) Plants and animals have mitochondria but no chloroplasts; fungi have only chloroplasts.

5. What are the typical periods in the unfolding of a behavioral act?
- (a) Appetitive behavior, exploratory behavior, quiescence
 - (b) Termination, appetitive behavior, exploratory behavior
 - (c) Appetitive behavior, consummatory act, quiescence
 - (d) Consummatory act, exploratory behavior, termination
 - (e) Appetitive behavior, termination, consummatory act

True / False (10 marks)

Answer True or False

- 6. True or False: Using negative reinforcement exclusively can accurately assess a dog's hearing sensitivity to various sound frequencies.
- 7. True or False: Down's syndrome is primarily caused by trisomy of chromosome 21, resulting in an extra copy of this chromosome.
- 8. True or False: Parenchyma and companion cells are primarily responsible for water conduction in plants.
- 9. True or False: Stabilizing selection leads to extreme traits becoming more common in a population, enhancing diversity.
- 10. True or False: The rise in intracellular calcium in the sea urchin oocyte triggers the acrosomal reaction, allowing multiple sperm to fertilize the egg.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the significance of courtship behaviors in the common tern, particularly the male's presentation of fish, and analyze how such behaviors facilitate reproductive success in avian species.
- 12. Discuss the essential characteristics that define living organisms and explain why a mere combination of chemical components cannot constitute life.

End of Paper